

Fourier, Delta Functions, and Probability

The uncertainty principle, three parts before quantum mechanics, as a theorem about integrals.

WHERE WE ARE

Part 0 closes here, and it closes with the one sentence that organises everything below: **the Fourier transform is a change of basis (Chapter 0.4) in an infinite-dimensional inner-product space (Chapter 0.5), and the basis is chosen because it is the one in which the derivative operator is diagonal.** That is the whole subject. Every formula in this chapter is a consequence of it, and if a formula ever looks arbitrary, go back to that sentence and read it again.

Chapter 0.5 proved that a Hermitian operator has an orthonormal basis of eigenvectors, and that in that basis it is nothing but a list of numbers. Chapter 0.8 found that $e^{\lambda t}$ are the eigenfunctions of d/dt . Put those two together — take the eigenfunctions of the derivative and use them as a basis — and you get Fourier analysis, whereupon every linear differential equation in this book becomes an algebra problem. Differentiation becomes multiplication. Convolution becomes multiplication. Green's functions become division.

And then the payoff, which is the reason this chapter exists. In §6 you will define the width of a function and the width of its transform, and *prove* that their product cannot be smaller than one half. That inequality is the Heisenberg uncertainty principle. You will have derived it with no particles, no measurement, no $\hbar$, and no physics of any kind — using Cauchy–Schwarz from Chapter 0.5 and integration by parts from Chapter 0.2. Chapter 4.6 will add exactly one thing to it: the substitution $p = \hbar k$.

TOOLS YOU'LL NEED — Chapter 0.2: integration by parts, the Gaussian integral $\int e^{-ax^2} dx = \sqrt{\pi/a}$ and its completed-square cousin, and Problem 4 (the transform of a Gaussian), which §6 cashes in explicitly. Chapter 0.3: Euler's formula, and Taylor expansion to second order with control of the remainder. Chapter 0.5: inner products, Cauchy–Schwarz with its equality condition, orthonormal bases and the coefficient formula $c_i = \langle e_i, v \rangle$, completeness $\sum_i |e_i\rangle \langle e_i| = \hat{I}$, unitarity, and Dirac notation. Chapter 0.7: the divergence theorem and the diffusion equation. Chapter 0.8: that $e^{\lambda t}$ is the eigenfunction of d/dt , the driven oscillator, and the Duhamel/integrating-factor solution. Sections 3 and 6 each cash a specific theorem from Chapters 0.2 and 0.5; nothing here is decoration.

1 • Fourier series as an orthonormal basis

Chapter 0.5's table of inner-product spaces had a row we skipped past: functions on an interval, with

$$\langle f, g \rangle \equiv \int_{-L/2}^{L/2} \overline{f(x)} g(x) dx. \quad (0.9.1)$$

Nothing new is being asserted. This is the same definition Chapter 0.5 gave, with the vectors now being functions: conjugate symmetry because $\overline{\int \bar{g} f} = \int \bar{f} g$, linearity in the second slot because integration is linear, positive definiteness because $\int |f|^2 \geq 0$ and vanishes only for $f \equiv 0$. Chapter 0.4 warned that a space of functions was coming and that it would be infinite-dimensional. Here it is.

Everything Chapter 0.5 proved about inner-product spaces is therefore available. What we do not yet have is a *basis*. We are about to choose one, and the choice is not aesthetic: §3 shows that the functions below are precisely the eigenfunctions of d/dx .

1.1 · The orthogonality relation, by direct integration

Work on the interval $[-L/2, L/2]$ and define the **wavenumbers**

$$k_n = \frac{2\pi n}{L}, \quad n \in \{\dots, -2, -1, 0, 1, 2, \dots\}. \quad (0.9.2)$$

These are exactly the k for which e^{ikx} returns to its starting value after one interval length: $e^{ik_n(x+L)} = e^{ik_n x} e^{2\pi i n} = e^{ik_n x}$, using Euler from Chapter 0.3. Now compute the inner product of two of them. For $m = n$ the integrand is $e^{-ik_n x} e^{ik_n x} = 1$ and the integral is L . For $m \neq n$, write $\kappa = k_n - k_m = 2\pi(n - m)/L$, which is not zero, and integrate directly:

$$\begin{aligned} \int_{-L/2}^{L/2} e^{i\kappa x} dx &= \left[\frac{e^{i\kappa x}}{i\kappa} \right]_{-L/2}^{L/2} = \frac{e^{i\kappa L/2} - e^{-i\kappa L/2}}{i\kappa} \\ &= \frac{2 \sin(\kappa L/2)}{\kappa} = \frac{2 \sin(\pi(n - m))}{\kappa} = 0, \end{aligned} \quad (0.9.3)$$

because $n - m$ is a non-zero integer and $\sin$ of an integer multiple of π vanishes. (The middle step used $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$, which is Euler's formula subtracted from its own conjugate.) Both cases together:

$$\boxed{\int_{-L/2}^{L/2} e^{-ik_m x} e^{ik_n x} dx = L \delta_{mn}} \quad (0.9.4)$$

That is the entire technical content of Fourier series. Divide by $\sqrt{L}$ to normalise and the functions

$$e_n(x) = \frac{1}{\sqrt{L}} e^{ik_n x} \quad \text{satisfy} \quad \langle e_m, e_n \rangle = \delta_{mn} \quad (0.9.5)$$

— an **orthonormal set** in exactly Chapter 0.5's sense $\langle e_i, e_j \rangle = \delta_{ij}$, verified by integration rather than assumed.

1.2 · Coefficients are inner products

Chapter 0.5 proved that in an orthonormal basis, finding coordinates requires no work at all: $v = \sum_i \langle e_i, v \rangle e_i$. Transcribe it. Suppose $f = \sum_n c_n e_n$. Take $\langle e_m, \cdot \rangle$ of both sides and use (0.9.5):

$$\langle e_m, f \rangle = \sum_n c_n \langle e_m, e_n \rangle = \sum_n c_n \delta_{mn} = c_m. \quad (0.9.6)$$

Not one integral had to be solved and not one linear system inverted; the orthogonality did the work. Writing it out and folding the $1/\sqrt{L}$ into the coefficient, the standard form is

$$f(x) = \sum_{n=-\infty}^{\infty} a_n e^{ik_n x}, \quad a_n = \frac{1}{L} \int_{-L/2}^{L/2} f(x) e^{-ik_n x} dx. \quad (0.9.7)$$

1.3 · Completeness — quoted, and flagged as quoted

► QUOTED, NOT DERIVED

Everything above shows that the e_n are *orthonormal*. It does not show that they are *enough* — that every function of interest is actually reachable as a sum (0.9.7). That is **completeness**, and we are going to use it without proving it.

The honest statement: for every f with $\int_{-L/2}^{L/2} |f|^2 dx < \infty$, the partial sums $S_N = \sum_{|n| \leq N} a_n e^{ik_n x}$ converge to f in the norm of (0.9.1), meaning $\|f - S_N\| \rightarrow 0$. This is L^2 convergence, and it is the version that matters for physics because it is the version that makes the energy bookkeeping work. It does *not* imply that $S_N(x) \rightarrow f(x)$ at every point — §1.4 exhibits a place where that fails permanently. Pointwise convergence needs extra hypotheses (the Dirichlet conditions: piecewise smooth, finitely many extrema and jumps), and at a jump the series converges to the *midpoint* of the two one-sided limits, not to either of them.

Both facts are proved in Chapter 4.3, where the space of square-integrable functions gets its proper name (a Hilbert space) and completeness stops being an assumption. Until then: we are standing on it, and you know that we are.

1.4 · A worked expansion, and an honest failure

Take $L = 2\pi$, so $k_n = n$, and expand the square wave

$$f(x) = \begin{cases} +1, & 0 < x < \pi, \\ -1, & -\pi < x < 0. \end{cases} \quad (0.9.8)$$

Since f is real and odd, work with sines directly: write $f = \sum_{n \geq 1} b_n \sin nx$, where $\int_{-\pi}^{\pi} \sin mx \sin nx dx = \pi \delta_{mn}$ (the same computation as (0.9.4), rewritten with Euler). The integrand $f(x) \sin nx$ is a product of two odd functions,

hence even, so the integral over $[-\pi, \pi]$ is twice the integral over $[0, \pi]$, where $f = +1$:

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx = \frac{2(1 - (-1)^n)}{\pi n}. \quad (0.9.9)$$

The bracket is 2 for odd n and 0 for even n , so

$$f(x) = \frac{4}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots \right). \quad (0.9.10)$$

A discontinuous function, built entirely out of smooth ones. That should bother you slightly, and it should: no finite partial sum is discontinuous, so the discontinuity is manufactured only in the limit — and the manufacturing process leaves a permanent scar.

The Gibbs phenomenon. Near the jump at $x = 0$, every partial sum overshoots. Not by less and less: by the *same amount*, forever. The partial sum S_M peaks at $1.178980 \dots$ rather than 1, and as M grows the overshoot narrows and slides toward the jump but never shrinks. Since the jump has size 2, the overshoot is 8.949% of the jump, and that number is a constant of nature for every jump discontinuity in every Fourier series. The grind box derives it.

This is a genuine failure of pointwise convergence, and it is worth naming as one rather than apologising for it. It is also not a failure of L^2 convergence: the overshoot region shrinks in width fast enough that $\int |f - S_M|^2 \rightarrow 0$ regardless. The two notions of convergence disagree, and the one physics needs is the one that survives.

▼ Grind box — where 1.178980 comes from

Take the partial sum of (0.9.10) to M terms (odd n only) and differentiate it — the sum of cosines telescopes into something closed:

$$S'_M(x) = \frac{4}{\pi} \sum_{n \text{ odd}}^M \cos nx = \frac{2 \sin(2Nx)}{\pi \sin x},$$

where $M = 2N - 1$; the closed form follows from summing the geometric series $\sum_{j=0}^{N-1} e^{i(2j+1)x}$ and taking real parts. Integrating from 0,

$$S_M(x) = \frac{2}{\pi} \int_0^x \frac{\sin(2Nt)}{\sin t} \, dt.$$

The first maximum is at the first zero of the integrand's numerator past the origin, i.e. $x^* = \pi/2N$. For large N that is small, so $\sin t \approx t$ throughout the range, and substituting $u = 2Nt$:

$$S_M(x^*) \rightarrow \frac{2}{\pi} \int_0^{\pi} \frac{\sin u}{u} \, du = \frac{2}{\pi} \text{Si}(\pi) = 1.178979 \dots$$

The number $\text{Si}(\pi) = 1.851937 \dots$ is not elementary — it is a value of the sine integral — but it is perfectly definite. Numerically, the peak of S_M is 1.181302 at $M = 11$, 1.179103 at $M = 51$,

1.178988 at $M = 201$, 1.178980 at $M = 1001$. It is converging, and not to 1.

Why it matters beyond aesthetics. Any time you band-limit a signal — truncate its Fourier content, which is what every detector, every lens and every digital filter does — you ring at the edges by 9% of the step. Sharp edges and finite bandwidth are incompatible, and §6 makes that a theorem.

2 · From series to transform

Fourier series need a box. Physics on the whole line does not have one, so we take $L \rightarrow \infty$ and watch what survives.

2.1 · The limit, taken carefully

The allowed wavenumbers (0.9.2) are spaced by

$$\Delta k = k_{n+1} - k_n = \frac{2\pi}{L}, \quad (0.9.11)$$

so as the box grows the modes crowd together and in the limit form a continuum. The sum over n in (0.9.7) is going to become an integral over k , but only if we first arrange for it to *look* like a Riemann sum — that is, if each term carries a factor of Δk . So define, for a function on the whole line,

$$\tilde{f}(k) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx \quad (0.9.12)$$

and notice that the coefficient in (0.9.7) is $a_n = \sqrt{2\pi} \tilde{f}(k_n) / L$. Substituting that back:

$$f(x) = \sum_n \frac{\sqrt{2\pi}}{L} \tilde{f}(k_n) e^{ik_n x} = \frac{1}{\sqrt{2\pi}} \sum_n \tilde{f}(k_n) e^{ik_n x} \underbrace{\frac{2\pi}{L}}_{\Delta k}. \quad (0.9.13)$$

The factor $2\pi/L$ that had to be manufactured is exactly the mode spacing (0.9.11). So the right-hand side is a Riemann sum, of the kind Chapter 0.2 defined the integral to be the limit of, and letting $L \rightarrow \infty$ gives

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk \quad (0.9.14)$$

The two boxed formulas are the **Fourier transform pair**. Compare them with Chapter 0.5's $v = \sum_i \langle e_i, v \rangle e_i$: (0.9.12) is the coefficient $\langle e_k, f \rangle$, and (0.9.14) is the expansion, with the sum over a discrete index replaced by an integral over a continuous one. It is the same equation. What has changed is that the basis is now labelled by a real number instead of an integer, and this is precisely the difficulty that Chapter 4.3 has to work to make legitimate: the functions e^{ikx} do

not belong to the space they are supposed to be a basis of, because $\int |e^{ikx}|^2 dx = \int 1 dx = \infty$. Chapter 0.5's warning box said this would happen. §5 gives it the tool it needs.

2.2 · The convention, chosen and held

The 2π has to go somewhere and there is no canonical place to put it. We have chosen the **symmetric convention**: a factor $1/\sqrt{2\pi}$ on each of (0.9.12) and (0.9.14), and the sign e^{-ikx} going forward, e^{+ikx} coming back. We will hold it for the entire book.

Other texts choose differently, and this is a real and recurring source of confusion — factors of 2π and $\sqrt{2\pi}$ appear and disappear between references for no mathematical reason whatsoever. The three you will meet:

CONVENTION	FORWARD	INVERSE	CONVOLUTION THEOREM
Symmetric (ours)	$\frac{1}{\sqrt{2\pi}} \int f e^{-ikx} dx$	$\frac{1}{\sqrt{2\pi}} \int \tilde{f} e^{ikx} dk$	$\sqrt{2\pi} \tilde{f} \tilde{g}$
Asymmetric	$\int f e^{-ikx} dx$	$\frac{1}{2\pi} \int \tilde{f} e^{ikx} dk$	$\tilde{f} \tilde{g}$
Ordinary frequency	$\int f e^{-2\pi i \nu x} dx$	$\int \tilde{f} e^{2\pi i \nu x} d\nu$	$\tilde{f} \tilde{g}$

The symmetric one is chosen here for one reason: it makes the transform *unitary*, which is the next result, and unitary is the property Chapter 4.4 needs. Note also that in time-dependent problems physics usually writes $e^{-i\omega t}$ for the *forward* direction in time, i.e. the opposite sign from our e^{-ikx} in space, so that a plane wave reads $e^{i(kx - \omega t)}$. That is a convention about which way waves travel, not a different transform; we will flag it when it first matters.

2.3 · Plancherel: the transform is a rotation

Chapter 0.5's Parseval identity said that in an orthonormal basis, the squared length of a vector is the sum of the squared magnitudes of its coordinates. Take that statement into the box and then to the limit.

In the box, expand $\int |f|^2$ using (0.9.7) and collapse the double sum with the orthogonality (0.9.4) — only the diagonal survives:

$$\int_{-L/2}^{L/2} |f|^2 dx = \sum_{m,n} \bar{a}_m a_n \int e^{i(k_n - k_m)x} dx = L \sum_n |a_n|^2. \quad (0.9.15)$$

Now substitute $a_n = \sqrt{2\pi} \tilde{f}(k_n)/L$ and manufacture the mode spacing again:

$$L \sum_n \frac{2\pi}{L^2} |\tilde{f}(k_n)|^2 = \sum_n |\tilde{f}(k_n)|^2 \frac{2\pi}{L} \xrightarrow{L \rightarrow \infty} \int_{-\infty}^{\infty} |\tilde{f}(k)|^2 dk. \quad (0.9.16)$$

So we have proved **Plancherel's theorem**:

$$\boxed{\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\tilde{f}(k)|^2 dk} \quad (0.9.17)$$

Read it structurally rather than as a formula. It says the map $f \mapsto \tilde{f}$ preserves norms. And a norm-preserving linear map on an inner-product space preserves the inner product too, because the inner product is recoverable from the norm — expand $\|f + g\|^2$ and $\|f + ig\|^2$ and solve. Hence

$$\langle f, g \rangle = \int \bar{f}g dx = \int \bar{\tilde{f}} \tilde{g} dk = \langle \tilde{f}, \tilde{g} \rangle. \quad (0.9.18)$$

THE FOURIER TRANSFORM IS UNITARY

(0.9.18) is exactly Chapter 0.5's $\langle Uu, Uv \rangle = \langle u, v \rangle$, the defining property of a **unitary** map. So the Fourier transform is not merely a useful formula. It is a *rotation in function space*: it preserves lengths, preserves angles, and carries orthonormal bases to orthonormal bases. Nothing is created or destroyed by transforming; the object is the same, viewed along different axes.

Chapter 0.5 also proved that unitary maps are exactly the ones that conserve total probability. So this one line is the reason a wavefunction in Chapter 4.4 can be moved freely between position and momentum representations without the probabilities ceasing to add to one. Physicists say the Fourier transform "conserves probability". They mean it is a rotation.

3 · Why Fourier, actually: the derivative becomes multiplication

Now the reason the subject exists. Everything so far could have been said about any orthonormal basis. This section says what is special about *this* one.

3.1 · The derivative theorem

Transform f' and integrate by parts, which by Chapter 0.2 is the product rule run backwards:

$$\begin{aligned} \tilde{f}'(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f'(x) e^{-ikx} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[f(x) e^{-ikx} \right]_{-\infty}^{\infty} + \frac{ik}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx. \end{aligned} \quad (0.9.19)$$

The boundary term is dropped, and the assumption behind dropping it should be said out loud: $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Since $|e^{-ikx}| = 1$ exactly, the oscillating factor cannot help; the decay must come from f itself. Physically this is the statement that the disturbance is localised — the wave packet is somewhere, the field dies away, the concentration is finite. It is a physical hypothesis wearing mathematical clothes, and it is the same hypothesis Chapter 0.2 flagged when it derived integration by parts. With it,

$$\boxed{\widetilde{f'}(k) = ik \widetilde{f}(k)} \quad (0.9.20)$$

and by iterating, $\widetilde{f^{(m)}} = (ik)^m \widetilde{f}$. **Differentiation has become multiplication.**

3.2 · Why: the basis diagonalises d/dx

Do not accept (0.9.20) as a computational trick. It is the spectral theorem. Chapter 0.8 observed that exponentials are the eigenfunctions of the derivative; write it in the form this chapter needs:

$$\frac{d}{dx} e^{ikx} = ik e^{ikx}. \quad (0.9.21)$$

The operator d/dx acting on the function e^{ikx} returns the same function times the number ik . That is Chapter 0.5's $Av = \lambda v$ with $A = d/dx$, $v = e^{ikx}$, $\lambda = ik$. So the Fourier basis is the eigenbasis of the derivative operator, and (0.9.20) is nothing but Chapter 0.5's observation that *an operator, written in its own eigenbasis, is a list of numbers* — here the list $\{ik\}$, indexed by a continuum.

THIS IS WHY FOURIER ANALYSIS EXISTS

A linear differential equation is an operator equation $\mathcal{L}y = F$, where $\mathcal{L}$ is built out of d/dx and multiplication by constants. In the Fourier basis every d/dx becomes ik , so $\mathcal{L}$ becomes a *polynomial in ik* — an ordinary function, one number for each k . The equation becomes

$$P(ik) \tilde{y}(k) = \tilde{F}(k) \quad \implies \quad \tilde{y}(k) = \frac{\tilde{F}(k)}{P(ik)}.$$

Solving a differential equation has been reduced to **dividing**. All the work has moved into transforming in and transforming back. This is exactly what diagonalising a matrix does for a linear system in Chapter 0.5, and it is the same theorem: a hard coupled problem becomes a list of trivial uncoupled ones, one per basis direction.

The honest caveat: Chapter 0.5's spectral theorem was proved in *finite* dimensions, and the argument above applies it to an operator on a function space, where the eigenfunctions are not even in the space. That gap is real. Chapter 4.3 closes it, and the price is a genuinely more careful theory. We proceed knowing what we are standing on.

3.3 · A demonstration: the driven oscillator

Chapter 0.8 fought the damped driven oscillator with trial solutions and complex amplitudes:

$$\ddot{y} + 2\gamma\dot{y} + \omega_0^2 y = F(t). \quad (0.9.22)$$

Transform in t with conjugate variable ω , using (0.9.20) twice — each time derivative becomes a factor $i\omega$:

$$(-\omega^2 + 2i\gamma\omega + \omega_0^2) \tilde{y}(\omega) = \tilde{F}(\omega). \quad (0.9.23)$$

The differential equation has become an algebraic equation, for each ω separately, and it is solved by division:

$$\tilde{y}(\omega) = \chi(\omega) \tilde{F}(\omega), \quad \chi(\omega) = \frac{1}{\omega_0^2 - \omega^2 + 2i\gamma\omega}. \quad (0.9.24)$$

The function χ is the **transfer function** (or response function, or susceptibility — three names for one object). Everything Chapter 0.8 extracted by hand is now readable off it. The response amplitude is $|\chi|$, which peaks where the denominator is smallest — resonance. The phase lag is $-\arg \chi$, which passes through $\pi/2$ at $\omega = \omega_0$. And near resonance the denominator is approximately $2\omega_0(\omega_0 - \omega) + 2i\gamma\omega_0$, so

$$|\chi|^2 \approx \frac{1}{4\omega_0^2} \cdot \frac{1}{(\omega_0 - \omega)^2 + \gamma^2}, \quad (0.9.25)$$

the **Lorentzian** of Chapter 0.8, with half-width-at-half-maximum γ , now derived rather than fitted. Problem 4 turns this same shape into the natural linewidth of an excited state, and the connection runs through §6.

▼ Grind box — what "transform the equation" is really doing, and two traps

It is a change of basis, and only that. In Chapter 0.5's language, an equation $\mathcal{L}y = F$ in the position basis becomes, after inserting completeness, an equation between components in the k basis. Since $\mathcal{L}$ is diagonal there, the matrix equation $\sum_j L_{ij}y_j = F_i$ degenerates to $L_{ii}y_i = F_i$ with no sum — one equation per mode, each solved by dividing. There is no new mathematics in §3 at all. There is a change of coordinates.

Trap 1: initial conditions. Transforming in t over the whole line quietly assumes the solution exists for all time and decays at both ends. That is the right thing to want for a *steady-state* response, and the wrong thing for an initial-value problem, where the system is switched on at $t = 0$ and you are told $y(0)$ and $\dot{y}(0)$. For those, the correct tool is the Laplace transform — the same idea with $i\omega \rightarrow s$ and the integral run over $[0, \infty)$, which produces the initial data as boundary terms in the integration by parts instead of discarding them. Everything in this chapter has a Laplace twin.

Trap 2: $P(ik)$ can vanish. Dividing by $P(ik)$ is illegal where P has a zero on the real axis, which is exactly the undamped resonance $\gamma = 0, \omega = \omega_0$. The division then fails because the answer genuinely does not exist: an undamped oscillator driven exactly at resonance has no bounded steady state, it grows linearly forever. The formula's failure is reporting a physical fact. When damping is present, P has no real zeros and the division is safe — which is why $\gamma > 0$ was doing more work in Chapter 0.8 than it looked.

4 · Convolution and Green's functions

Definition. The **convolution** of two functions is

$$(f * g)(x) \equiv \int_{-\infty}^{\infty} f(y) g(x - y) dy. \quad (0.9.26)$$

Read it as: *at each point x , take a weighted average of f , with the weights given by g centred on x and reflected.* It is symmetric — substituting $u = x - y$ turns (0.9.26) into $\int g(u) f(x - u) du$, so $f * g = g * f$ — and it is bilinear, so it is a genuine product on the space of functions.

4.1 · The convolution theorem, derived

Transform (0.9.26) and split the exponential using $x = y + (x - y)$:

$$\begin{aligned} \widetilde{f * g}(k) &= \frac{1}{\sqrt{2\pi}} \iint f(y) g(x - y) e^{-ikx} dy dx \\ &= \frac{1}{\sqrt{2\pi}} \iint f(y) e^{-iky} g(u) e^{-iku} dy du, \end{aligned} \quad (0.9.27)$$

where the second line substituted $u = x - y$ at fixed y — a shift, so $du = dx$ and the limits are unchanged. The double integral now factorises, because the integrand is a product of a function of y alone and a function of u alone:

$$\boxed{\widetilde{f * g}(k) = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} \tilde{f}(k) \cdot \sqrt{2\pi} \tilde{g}(k) = \sqrt{2\pi} \tilde{f}(k) \tilde{g}(k)} \quad (0.9.28)$$

(The stray $\sqrt{2\pi}$ is the price of the symmetric convention; the table in §2.2 lists what it becomes in the others. Get it wrong and every amplitude in Part V is off by a constant factor, so it is worth one moment of care.) **Convolution in x is multiplication in k .** An operation that costs a full integral at every single point becomes a pointwise product.

4.2 · Linear time-invariant systems, and Green's functions

Here is why (0.9.28) is one of the most load-bearing facts in physics. Consider any system that is **linear** (double the input, double the output; inputs superpose) and **time-invariant** (its behaviour does not depend on what o'clock it is). Feed it an input $F(t)$ and call the output $y(t)$.

Let $G(t)$ be the output produced by a single sharp unit impulse at $t = 0$ — the **impulse response**, or **Green's function**. Time-invariance says an impulse at t' produces $G(t - t')$. Linearity says a general input, regarded as a dense sequence of impulses of strength $F(t') dt'$, produces the superposition of the individual responses:

$$y(t) = \int_{-\infty}^{\infty} G(t - t') F(t') dt' = (G * F)(t). \quad (0.9.29)$$

That is the whole content of linear response theory, and it is a convolution. Transform it with (0.9.28) and it becomes $\tilde{y} = \sqrt{2\pi} \tilde{G} \tilde{F}$ — compare (0.9.24) and read off that the transfer function is $\chi = \sqrt{2\pi} \tilde{G}$. The transfer function is the transformed Green's function. Two objects, one thing.

Chapter 0.8 already built one of these and said so. Its integrating-factor solution of $\dot{y} + \lambda y = F(t)$ (Chapter 0.8 calls the rate constant k ; here k is taken, so it is λ) was

$$y(t) = \int_{-\infty}^t e^{-\lambda(t-t')} F(t') dt' = \int_{-\infty}^{\infty} G(t-t') F(t') dt', \quad (0.9.30)$$

with $G(u) = e^{-\lambda u}$ for $u > 0$ and $G(u) = 0$ for $u < 0$. The upper limit t in Chapter 0.8's formula and the vanishing of G for negative argument are the same statement, and that statement is **causality**: the response cannot precede the impulse. Duhamel's formula was a convolution all along.

WHERE THIS GOES: FEYNMAN DIAGRAMS ARE PRODUCTS BECAUSE CONVOLUTION IS MULTIPLICATION

Everything above generalises with t replaced by a spacetime point. In Chapter 5.6 the operator $\mathcal{L}$ is a wave operator, $\square + m^2$, and its Green's function is called the **propagator** $\Delta(x-y)$ — the amplitude for a disturbance created at y to arrive at x . Transforming, the propagator becomes

$$\tilde{\Delta}(p) = \frac{1}{-p^2 + m^2} \quad (\text{up to conventions and an } i\epsilon),$$

which is (0.9.24) with $\omega \rightarrow p$: division by the polynomial, exactly as in §3.

Now the payoff. A particle propagating from a to b , interacting, then propagating to c contributes an amplitude that is a *convolution* of propagators in position space — an integral over the unknown interaction point. By (0.9.28), in momentum space that integral collapses into a **product**. This is why the Feynman rules read the way they do: *write down a factor for each line and each vertex and multiply them together*. There is no deeper combinatorial magic in the rules. Feynman diagrams are products in momentum space because the underlying operation is convolution in position space, and the convolution theorem is (0.9.28), which you have just proved in four lines. That is a real and load-bearing insight, and it is worth knowing now rather than in Part V.

5 • The Dirac delta, handled honestly

§4 needed "a single sharp unit impulse" and did not define one. §2 needed a completeness relation for a continuum of basis functions and did not have one. The same object supplies both, and it is not a function.

5.1 • Define it by what it does

Do not ask what value $\delta(x)$ takes at each point; there is no answer, and §7's warning box explains why the question is malformed. Ask instead what it *does* to other functions. The **Dirac delta** is defined by

$$\boxed{\int_{-\infty}^{\infty} \delta(x) f(x) dx = f(0)} \quad (0.9.31)$$

for every sufficiently well-behaved *test function* f . That is the definition, complete. Note what kind of object it makes δ : a rule that eats a function and returns a number, and does so linearly, since $\int \delta(\alpha f + \beta g) = \alpha f(0) + \beta g(0)$. A linear map from functions to numbers is a **linear functional** — precisely Chapter 0.5's definition of a *bra*. Objects defined this way are called **distributions**, and δ is the simplest non-trivial one.

Shifting the argument gives the general **sifting property**, by substituting $u = x - a$:

$$\int_{-\infty}^{\infty} \delta(x - a) f(x) dx = f(a). \quad (0.9.32)$$

The delta reaches into a function and pulls out its value at one point, discarding everything else. Every use of δ in this book is that sentence.

5.2 · It is the limit Chapter 0.2 was drawing

Take the normalised Gaussian, whose total area is 1 by Chapter 0.2's $\int e^{-ax^2} dx = \sqrt{\pi/a}$:

$$\delta_{\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}, \quad \int_{-\infty}^{\infty} \delta_{\sigma}(x) dx = 1 \text{ for every } \sigma. \quad (0.9.33)$$

This is *exactly* the curve in Chapter 0.2's width figure — the one whose measured area stayed pinned at 1.00000 while the peak rose as $1/\sigma$, and whose caption said that pushing σ to its minimum was watching the Dirac delta form. It was. Now the claim, made precise: for continuous f ,

$$\lim_{\sigma \rightarrow 0} \int_{-\infty}^{\infty} \delta_{\sigma}(x) f(x) dx = f(0). \quad (0.9.34)$$

Proof. Since $\int \delta_{\sigma} = 1$, we may write the difference as

$$\int \delta_{\sigma}(x) [f(x) - f(0)] dx. \quad (0.9.35)$$

Fix $\varepsilon > 0$. By continuity there is an $\eta > 0$ with $|f(x) - f(0)| < \varepsilon$ for $|x| < \eta$; that part of the integral is therefore smaller than $\varepsilon \int \delta_{\sigma} = \varepsilon$. Outside $|x| < \eta$, the factor δ_{σ} is at most $e^{-\eta^2/2\sigma^2}/(\sigma\sqrt{2\pi})$, which goes to zero faster than any power as $\sigma \rightarrow 0$, so (for bounded f) that part vanishes in the limit. Hence the whole thing is below 2ε for small enough σ , and ε was arbitrary. ■

Two things are worth extracting. First, the limit is taken *after* integrating, never before: $\lim_{\sigma} \delta_{\sigma}(x)$ pointwise is 0 for $x \neq 0$ and $+\infty$ at $x = 0$, which is useless. The delta lives inside the integral sign and nowhere else. Second, the Gaussian was not special — any normalised bump that narrows works, and Chapter 0.2's grind box on differentiating

under the integral sign exhibited another one ($2xe^{-x^2/a^2}/a^2$, on the half-line) and called it "the Dirac delta being born". It was.

5.3 · The Fourier representation, and what it really is

The single most-used identity in Parts IV–VII is

$$\boxed{\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk} \quad (0.9.36)$$

which cannot be true as it stands, since the integrand has modulus 1 everywhere and the integral does not converge. Here is the derivation that makes it respectable, and it uses only Chapter 0.2. Insert a convergence factor $e^{-\epsilon k^2}$ with $\epsilon > 0$, do the now-honest Gaussian integral with $b = ix$, and take $\epsilon \rightarrow 0$ at the end:

$$\begin{aligned} \frac{1}{2\pi} \int e^{ikx} e^{-\epsilon k^2} dk &= \frac{1}{2\pi} \sqrt{\frac{\pi}{\epsilon}} e^{-x^2/4\epsilon} \\ &= \frac{1}{\sqrt{4\pi\epsilon}} e^{-x^2/4\epsilon} = \delta_\sigma(x) \Big|_{\sigma=\sqrt{2\epsilon}}. \end{aligned} \quad (0.9.37)$$

The regulated integral is a normalised Gaussian of width $\sqrt{2\epsilon}$ — precisely (0.9.33) — which tends to δ by §5.2. So (0.9.36) is true in the only sense in which it was ever meant: *as a limit of honest integrals, evaluated under an integral sign*. That manoeuvre — insert a regulator, compute, remove the regulator — is not a dodge to be embarrassed about. It is the same move that will be called *renormalisation* in Chapter 5.9, met here in its harmless form.

Now see what (0.9.36) actually is. Chapter 0.5's completeness relation reads $\sum_n |e_n\rangle \langle e_n| = \hat{I}$; in function language, with $e_n(x)$ the components, that says $\sum_n e_n(x) \overline{e_n(y)} = \delta(x - y)$, since acting on f it must return f . Put in the Fourier basis (0.9.5), $e_n = e^{ik_n x} / \sqrt{L}$, and manufacture the mode spacing one last time:

$$\sum_n \frac{e^{ik_n(x-y)}}{L} = \frac{1}{2\pi} \sum_n e^{ik_n(x-y)} \Delta k \xrightarrow{L \rightarrow \infty} \frac{1}{2\pi} \int e^{ik(x-y)} dk. \quad (0.9.38)$$

So (0.9.36) is **the completeness relation $\sum_n |e_n\rangle \langle e_n| = \hat{I}$ in continuous disguise**. That is the honest reading, and it explains why the delta appears everywhere the Fourier transform does: every time you insert the identity in a continuous basis, a delta drops out. In Dirac notation the whole of §2 compresses to $\langle x|k\rangle = e^{ikx} / \sqrt{2\pi}$ and $\int |k\rangle \langle k| dk = \hat{I}$, and (0.9.36) is $\langle x|y\rangle = \delta(x - y)$.

5.4 · The rules, derived

Scaling. Claim: $\delta(ax) = \delta(x) / |a|$ for $a \neq 0$. Test it against an arbitrary f and substitute $u = ax$. If $a > 0$ then $dx = du/a$ and the limits keep their order:

$$\int \delta(ax) f(x) dx = \int \delta(u) f(u/a) \frac{du}{a} = \frac{f(0)}{a}. \quad (0.9.39)$$

If $a < 0$ the substitution reverses the limits, and swapping them back supplies a minus sign, so the answer is $-f(0)/a = f(0)/|a|$. Both cases give $f(0)/|a|$, which is what $\int [\delta(x)/|a|] f(x) dx$ gives. Since the two functionals agree on every test function, they are equal:

$$\delta(ax) = \frac{\delta(x)}{|a|}. \quad (0.9.40)$$

Note the consequence: δ is not dimensionless. If x carries units of length then $\delta(x)$ carries units of $1/\text{length}$, so that (0.9.31) balances. Checking that in an unfamiliar formula catches an impressive number of errors.

The derivative. We would like δ' to obey integration by parts. So *define* it to, discarding the boundary term (test functions vanish at infinity):

$$\int \delta'(x) f(x) dx \equiv - \int \delta(x) f'(x) dx = -f'(0). \quad (0.9.41)$$

This is how *every* distribution is differentiated, and it is why distributions are so useful: they can always be differentiated, as many times as you like, with no smoothness required of anything. The step function has a derivative — it is δ — and δ has a derivative, and so on forever. Chapter 5.2's field equations lean on this constantly.

The three-dimensional delta, and Chapter 0.7's loose end. Define $\delta^3(\mathbf{r}) = \delta(x)\delta(y)\delta(z)$, so that $\int \delta^3(\mathbf{r}) f(\mathbf{r}) d^3r = f(\mathbf{0})$. Chapter 0.7 left a warning: the Coulomb potential $1/r$ satisfies $\nabla^2(1/r) = 0$ everywhere the computation is legal, yet a point charge is manifestly a source. Both halves are true and the resolution is a distribution. Direct differentiation away from the origin gives $\nabla^2(1/r) = 0$ for $r \neq 0$. But integrate over a ball of any radius R and use the divergence theorem of Chapter 0.7, with $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$:

$$\int_{r < R} \nabla^2 \frac{1}{r} d^3r = \oint_{r=R} \nabla \frac{1}{r} \cdot d\mathbf{S} = -\frac{1}{R^2} \oint_{r=R} dS = -4\pi. \quad (0.9.42)$$

An object that vanishes everywhere except one point yet integrates to -4π over every ball containing that point is, by (0.9.31), exactly -4π times a delta:

$$\nabla^2 \frac{1}{r} = -4\pi \delta^3(\mathbf{r}). \quad (0.9.43)$$

Chapter 0.7's loose end is tied. And note what (0.9.43) says in the language of §4: $-1/4\pi r$ is the *Green's function of the Laplacian* — the response to a unit point source. Poisson's equation $\nabla^2 \phi = -\rho/\epsilon_0$ is therefore solved by convolution with it, which is

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r' \quad (0.9.44)$$

— Coulomb's law, recovered as a special case of §4.2 rather than posited.

▼ Grind box — distributions, said properly

The construction that makes all of §5 rigorous is due to Schwartz and is worth seeing once, in outline, because it explains exactly which manipulations are safe.

Test functions. Fix a space $\mathcal{D}$ of very well-behaved functions — infinitely differentiable, and either compactly supported or (the Schwartz space $\mathcal{S}$) decaying faster than any power along with all their derivatives. $\mathcal{S}$ is the natural home for Fourier analysis, because (0.9.20) and its partner $\widetilde{xf} = i d\tilde{f}/dk$ trade decay for smoothness in both directions, so the transform maps $\mathcal{S}$ onto itself.

Distributions. A distribution is a continuous linear map $T : \mathcal{D} \rightarrow \mathbb{C}$. Ordinary functions embed: g becomes $T_g[f] = \int gf$. Some distributions, like δ , are not of that form for any g — which is the precise sense of “ δ is not a function”.

Everything is defined by moving the operation onto the test function. Derivative: $T'[f] \equiv -T[f']$. Scaling and translation: as in §5.4. Fourier transform: $\tilde{T}[f] \equiv T[\tilde{f}]$ — which, applied to δ , gives $\tilde{\delta}[f] = \delta[\tilde{f}] = \tilde{f}(0) = \frac{1}{\sqrt{2\pi}} \int f$, so $\tilde{\delta} = 1/\sqrt{2\pi}$, a constant. A perfectly sharp spike in x has perfectly flat content in k . Hold that thought for §6.

What is not defined. There is no general product of two distributions. The recipe above works because differentiating or transforming a test function gives another test function; *multiplying* two distributions has no such move available, and δ^2 is genuinely undefined — not hard, not subtle, undefined. See the warning box.

6 · The bandwidth theorem

This is what the chapter was for.

6.1 · Defining a width

Let f be normalised, $\int |f|^2 dx = 1$, which by Plancherel (0.9.17) means $\int |\tilde{f}|^2 dk = 1$ as well. Then $|f|^2$ and $|\tilde{f}|^2$ are both non-negative and both integrate to one: they are probability densities, whether or not anybody intends them as such. So give each the standard deviation of §7:

$$(\Delta x)^2 = \int (x - \langle x \rangle)^2 |f|^2 dx, \quad (\Delta k)^2 = \int (k - \langle k \rangle)^2 |\tilde{f}|^2 dk, \quad (0.9.45)$$

with $\langle x \rangle = \int x |f|^2 dx$ and $\langle k \rangle = \int k |\tilde{f}|^2 dk$.

We may assume $\langle x \rangle = \langle k \rangle = 0$. Not as a convenience — as a fact, derived in two lines. Replacing $f(x)$ by $f(x + x_0)$ shifts $|f|^2$ bodily and changes $\tilde{f}$ only by the phase e^{ikx_0} , so Δx and Δk are both untouched while $\langle x \rangle$ moves by

$-x_0$. Replacing $f(x)$ by $e^{-ik_0 x} f(x)$ leaves $|f|^2$ alone and shifts $\tilde{f}(k)$ to $\tilde{f}(k + k_0)$, moving $\langle k \rangle$ by $-k_0$ with Δk untouched. So both means can be driven to zero without changing either spread, and we do so.

6.2 · The move that makes it work

The difficulty is that Δx is an integral over x and Δk is an integral over k ; they live in different places and there is nothing to compare. §2 and §3 together fix that. Use the derivative theorem (0.9.20) and then Plancherel (0.9.17):

$$(\Delta k)^2 = \int k^2 |\tilde{f}|^2 dk = \int |ik \tilde{f}|^2 dk = \int |\tilde{f}'|^2 dk = \int |f'|^2 dx. \quad (0.9.46)$$

Read that carefully, because it is the crux. **The spread in k has become a statement about f alone: it is the size of the derivative of f .** Which is obvious in hindsight — a function with a lot of high- k content is a function that wiggles, and a function that wiggles has a big derivative — but now it is an identity, not an intuition. Both integrals are now over x and we may compare them.

6.3 · Cauchy–Schwarz

Apply Chapter 0.5's inequality $|\langle u, v \rangle|^2 \leq \langle u, u \rangle \langle v, v \rangle$ in the function space with the inner product (0.9.1), to the two vectors

$$u = x f(x), \quad v = f'(x). \quad (0.9.47)$$

Their norms are exactly the two quantities we want, by (0.9.45) and (0.9.46):

$$\left| \int \overline{x f} f' dx \right|^2 \leq \underbrace{\int x^2 |f|^2 dx}_{(\Delta x)^2} \underbrace{\int |f'|^2 dx}_{(\Delta k)^2}. \quad (0.9.48)$$

So the theorem is proved as soon as we can put a lower bound on the left-hand side. Call it $I = \int x \bar{f} f' dx$ (the x is real, so the conjugation does nothing to it) and add I to its own conjugate. The two terms assemble into a derivative by the product rule:

$$I + \bar{I} = \int x (\bar{f} f' + f \bar{f}') dx = \int x \frac{d}{dx} |f|^2 dx. \quad (0.9.49)$$

Now integrate by parts (Chapter 0.2), with the boundary term dropped because $x |f(x)|^2 \rightarrow 0$ at both ends — a hypothesis, justified in the grind box:

$$I + \bar{I} = \left[x |f|^2 \right]_{-\infty}^{\infty} - \int |f|^2 dx = 0 - 1 = -1. \quad (0.9.50)$$

That is the whole calculation. The normalisation $\int |f|^2 = 1$, which looked like tidiness, has turned into the number on the right-hand side.

THE STEP YOU JUST DID IS THE COMMUTATOR

Write $\hat{X}$ for "multiply by x " and $\hat{D}$ for d/dx . Then for any f ,

$$[\hat{X}, \hat{D}]f = xf' - (xf)' = xf' - f - xf' = -f, \quad \text{so} \quad [\hat{X}, \hat{D}] = -\hat{I}.$$

And $\langle f, \hat{X}\hat{D}f \rangle = I$, while $\langle f, \hat{D}\hat{X}f \rangle = \int \bar{f}(xf)' = -\int \bar{f}'xf = -\bar{I}$ by the same integration by parts. So (0.9.50) reads $\langle f, [\hat{X}, \hat{D}]f \rangle = I + \bar{I} = -\|f\|^2 = -1$. The integration by parts and the commutator are the same computation, seen from two sides — which is exactly why Chapter 4.6 will be able to state the general uncertainty relation as $\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$, as Chapter 0.5's insight box already promised. The -1 here is the $[\hat{x}, \hat{p}] = i\hbar$ of quantum mechanics with the $\hbar$ not yet installed.

6.4 • The theorem

A complex number is at least as big as its real part in absolute value, so from (0.9.50),

$$|I|^2 \geq (\operatorname{Re} I)^2 = \left(\frac{I + \bar{I}}{2} \right)^2 = \frac{1}{4}. \quad (0.9.51)$$

Feed that into (0.9.48) and take the square root of both sides:

$$\boxed{\Delta x \Delta k \geq \frac{1}{2}} \quad (0.9.52)$$

Done. Line by line, the ingredients were: the definition of the transform (§2), Plancherel (§2), the derivative theorem (§3), Cauchy–Schwarz (Chapter 0.5), integration by parts (Chapter 0.2), and the normalisation. There is nothing else in it.

6.5 • The Gaussian saturates it

Chapter 0.5 recorded the equality condition for Cauchy–Schwarz: equality holds exactly when the two vectors are parallel. Here that reads $f' = \lambda x f$ for some constant λ . We also need $|I| = |\operatorname{Re} I|$ in (0.9.51), i.e. I real; and $I = \lambda \int x^2 |f|^2 = \lambda (\Delta x)^2$, so λ must be real. Solving $f'/f = \lambda x$ by separation (Chapter 0.8) gives $\ln f = \frac{1}{2}\lambda x^2 + \text{const}$, so

$$f(x) = C e^{\lambda x^2/2}, \quad \lambda < 0 \text{ for } f \text{ to be normalisable.} \quad (0.9.53)$$

The minimisers are exactly the Gaussians, and nothing else. Not "Gaussians are one example" — the derivation ran backwards from the equality condition and produced them uniquely.

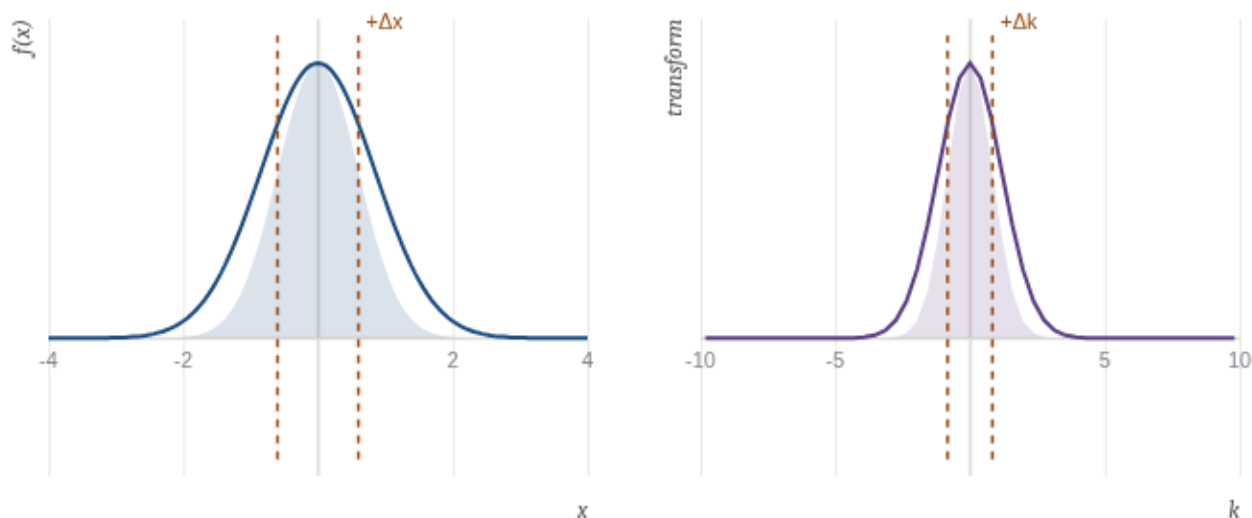
Now verify the value directly, cashing Chapter 0.2's Problem 4. Write $\lambda = -1/2\sigma^2$, so $f = Ce^{-x^2/4\sigma^2}$ and $|f|^2 \propto e^{-x^2/2\sigma^2}$, which by Chapter 0.2's Worked Example 2 has $\Delta x = \sigma$. Its transform, by $\int e^{-ax^2} e^{-ikx} dx = \sqrt{\pi/a} e^{-k^2/4a}$ with $a = 1/4\sigma^2$, is

$$\tilde{f}(k) = \frac{C}{\sqrt{2\pi}} \sqrt{\frac{\pi}{a}} e^{-k^2/4a} = C\sigma\sqrt{2} e^{-\sigma^2 k^2}, \quad (0.9.54)$$

so $|\tilde{f}|^2 \propto e^{-2\sigma^2 k^2} = e^{-k^2/2(1/2\sigma)^2}$, giving $\Delta k = 1/2\sigma$. Therefore

$$\Delta x \Delta k = \sigma \cdot \frac{1}{2\sigma} = \frac{1}{2} \quad \text{for every } \sigma. \quad (0.9.55)$$

The bound is attained, so (0.9.52) cannot be improved. Squeeze the Gaussian and its transform broadens by exactly the reciprocal factor; the product does not move. Which is a thing you should now watch happen.



shape **Gaussian** square pulse two humps width w = 0.60

$\Delta x = 0.600$ $\Delta k = 0.833$ $\Delta x \cdot \Delta k = 0.500$

Gaussian: the bound is attained. Drive the slider — the panels change, the product does not.

The bound, measured rather than asserted. Left: $f(x)$ (solid) with $|f|^2$ shaded; right: its transform $\tilde{f}(k)$, computed by an honest $O(N^2)$ discrete Fourier transform of the plotted samples, not from any formula. Dashed verticals mark $\pm\Delta x$ and $\pm\Delta k$. Both curves are scaled to unit peak, so only their *widths* are being compared. The three readouts are numerical second moments of the two shaded densities — the same arithmetic you would do on a data set. **Drive the width slider with the Gaussian selected.** The left panel narrows, the right panel visibly fattens, and $\Delta x \Delta k$ sits at 0.500 and refuses to move: that is (0.9.55) being measured. Now try the other shapes. **Two humps** gives a fringed transform and a product pinned at 1.493 — also constant, because the product depends on the *shape* and not the scale, but three times the floor. **Square pulse** gives the sinc, whose ringing tails fall off only as $1/k$; those tails carry so much weight at large k that $\int k^2 |\tilde{f}|^2 dk$ genuinely diverges, and the number you see drifts with the sampling grid rather than converging. That is not a bug in the figure. It is (0.9.52) being satisfied in the loudest available way, and it is Problem 1.

▼ Grind box — the hypotheses, said out loud

The boundary term. We assumed $x |f(x)|^2 \rightarrow 0$ at $\pm\infty$. Here is why it is automatic whenever the theorem's quantities are finite. By Cauchy–Schwarz, $\int |x| |f|^2 \leq (\int x^2 |f|^2)^{1/2} (\int |f|^2)^{1/2} = \Delta x < \infty$, so $x |f|^2$ is integrable. Its derivative is $|f|^2 + 2x \operatorname{Re}(\tilde{f} f')$, whose second piece is bounded in integral by $2\Delta x \Delta k < \infty$ — also integrable. A function whose derivative is integrable has limits at $\pm\infty$; a function that is itself integrable and has limits must have those limits equal to zero. So the boundary term vanishes for exactly the functions for which the statement has content.

When the spreads are infinite. If $\Delta x = \infty$ or $\Delta k = \infty$ the inequality is vacuously true, and this is not a rare corner: any function with a jump discontinuity has $\Delta k = \infty$ (Problem 1), and any function with power-law tails $|f|^2 \sim |x|^{-3}$ has $\Delta x = \infty$. The theorem is sharp only among functions smooth enough and localised enough to have both moments, and outside that class one uses different width

measures — the width at half maximum, the width containing 90% of the power, the reciprocal of the peak height. All obey inequalities of the same shape with different constants, and none of them is $\frac{1}{2}$.

Why the standard deviation, then? Because it is the measure for which the constant is exactly $\frac{1}{2}$ and the minimiser is exactly the Gaussian, and because it is the one that Chapter 4.6 needs — where $(\Delta x)^2$ is a variance of measurement outcomes and has to be that and nothing else.

YOU HAVE JUST PROVED THE UNCERTAINTY PRINCIPLE

(0.9.52) is the Heisenberg uncertainty principle. It is a theorem about Fourier transforms, and it has just been proved with no physics anywhere in it — no particles, no measurement, no observer, no $\hbar$. Every ingredient came from Part 0.

All that quantum mechanics will add, in Chapter 4.6, is a single substitution: $p = \hbar k$. That is one physical claim — the de Broglie relation — and it is the only new thing. Multiply both sides of (0.9.52) by $\hbar$:

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

The inequality was never quantum. It is a fact about waves, as true of a radio pulse or a seismic trace as of an electron, and it was known to signal engineers before it was known to physicists. What is quantum — what is genuinely strange and genuinely new — is the claim that a particle's momentum is the wavenumber of a wave. Once you accept that, the uncertainty principle is not an additional mystery. It is arithmetic you did in Part 0.

⚠ WHAT THE UNCERTAINTY PRINCIPLE DOES NOT SAY

It does not say that measuring a particle's position disturbs its momentum. That story — the microscope, the photon kicking the electron — is Heisenberg's own 1927 heuristic, it is not what the theorem states, and it has misled essentially every popular account since.

Look at what was actually proved. (0.9.52) is a relation between two **representations of one object**: the function f and its transform $\tilde{f}$ are the same thing written in two bases, and the theorem says their widths cannot both be small. No measurement occurs anywhere in the derivation. No observer appears. Nothing is disturbed, because nothing is done. A short pulse simply *is* broad in frequency — a 10 femtosecond flash of light contains a hundred nanometres of spectrum whether or not anyone looks at it (Problem 4), and a bass note plucked for a tenth of a second has an intrinsically fuzzy pitch. The spread is a property of the object, not of the act of examining it.

Measurement disturbance is a real and separate phenomenon with its own theorems, and Chapter 4.6 will keep the two apart. If you carry one thing away from §6, carry this: the uncertainty principle is a statement about what a localised wave *is*, not about what happens when you poke it.

7 · Probability, and why it is the same subject

Probability enters this chapter for a structural reason and not a curricular one: the central limit theorem is a Fourier argument, and the object that makes it one is a Fourier transform under another name.

7.1 · The apparatus, briefly

A **random variable** X taking real values is described by a **probability density** $p(x) \geq 0$ with $\int p = 1$, meaning $\Pr(a \leq X \leq b) = \int_a^b p$. Its **cumulative distribution** is $F(x) = \int_{-\infty}^x p(u) du$, and by the fundamental theorem of calculus (Chapter 0.2) $F' = p$: the density is the derivative of the CDF, which is the sentence that makes both objects worth having. The **expectation** of any function is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) p(x) dx, \quad (0.9.56)$$

which is linear in g because integration is. The **mean** is $\mu = \mathbb{E}[X]$, the **variance** is $\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$, and expanding the square with linearity gives the form you actually compute with:

$$\text{Var}(X) = \mathbb{E}[X^2] - 2\mu \mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2. \quad (0.9.57)$$

The **standard deviation** is $\sigma = \sqrt{\text{Var}}$ — the same quantity §6 called a spread, computed from the same integral. The **Gaussian** density is

$$p(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (0.9.58)$$

whose normalising constant is Chapter 0.2's $\int e^{-ax^2} dx = \sqrt{\pi/a}$ with $a = 1/2\sigma^2$, and whose variance is σ^2 by Chapter 0.2's Worked Example 2. Nothing in (0.9.58) is quoted.

7.2 · The characteristic function is a Fourier transform

Define the **characteristic function** of X :

$$\varphi_X(t) \equiv \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} p(x) e^{itx} dx = \sqrt{2\pi} \tilde{p}(-t). \quad (0.9.59)$$

It is the Fourier transform of the density, up to the convention's constant and a sign. Three properties, each immediate. It always exists, because $|e^{itx}| = 1$ and p is integrable — unlike moments, which can be infinite. It determines p uniquely, because the transform is invertible by (0.9.14). And its Taylor coefficients are the moments: differentiating under the integral (Chapter 0.2) brings down factors of ix , so

$$\varphi_X(t) = 1 + i\mu t - \frac{1}{2}\mathbb{E}[X^2] t^2 + O(t^3). \quad (0.9.60)$$

7.3 · Independent sums convolve, so characteristic functions multiply

Let X and Y be independent, with densities p_X, p_Y , and set $Z = X + Y$. Independence means the joint density factorises, $p(x, y) = p_X(x)p_Y(y)$. Then

$$F_Z(z) = \Pr(X + Y \leq z) = \int_{-\infty}^{\infty} p_X(x) \left[\int_{-\infty}^{z-x} p_Y(y) dy \right] dx, \quad (0.9.61)$$

and differentiating with respect to z under the integral, using $F' = p$ on the inner integral:

$$p_Z(z) = \int_{-\infty}^{\infty} p_X(x) p_Y(z - x) dx = (p_X * p_Y)(z). \quad (0.9.62)$$

Adding independent random variables convolves their densities. That is a non-obvious fact given a one-line derivation, and it is the bridge. Because now §4's convolution theorem applies. Transform (0.9.62), and translate back into characteristic functions with (0.9.59); the two stray factors of $\sqrt{2\pi}$ cancel against the one in (0.9.28):

$$\boxed{\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t)} \quad (0.9.63)$$

A hard operation on densities — an integral at every point — has become multiplication of two ordinary functions. This is (0.9.28) and nothing more, and it is the entire reason the central limit theorem is provable.

7.4 · Variance adds, and the $1/\sqrt{n}$

Take $\mu_X = \mu_Y = 0$ without loss of generality (subtract the means). Then

$$\mathbb{E}[(X + Y)^2] = \mathbb{E}[X^2] + 2 \mathbb{E}[XY] + \mathbb{E}[Y^2], \quad (0.9.64)$$

and the cross term factorises under independence, $\mathbb{E}[XY] = \iint xy p_X(x) p_Y(y) = \mathbb{E}[X] \mathbb{E}[Y] = 0$. Hence

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad (X, Y \text{ independent}). \quad (0.9.65)$$

Variances add; standard deviations do not. Everything below is bookkeeping on that one line. For n independent copies each of variance σ^2 , the sum has variance $n\sigma^2$; and since $\text{Var}(cX) = c^2 \text{Var}(X)$ directly from the definition, the mean $\bar{X}_n = \frac{1}{n} \sum X_i$ has

$$\text{Var}(\bar{X}_n) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}, \quad \text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}. \quad (0.9.66)$$

The **standard error** falls as $1/\sqrt{n}$, and the exponent $\frac{1}{2}$ is not a convention or an empirical rule. It is the square root of "variances add", which is (0.9.65), which is the vanishing of a cross term.

7.5 · The central limit theorem, sketched honestly

► A SKETCH, NOT A PROOF

What follows gets the mechanism exactly right and is genuinely where the Gaussian comes from. It leaves two things unproved, and both are named below rather than hidden.

Let $X_1, X_2, \dots$ be independent with the same distribution, mean μ , finite variance σ^2 . Standardise:

$$Z_n = \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n (X_i - \mu). \quad (0.9.67)$$

Write $Y = X - \mu$, so $\mathbb{E}[Y] = 0$ and $\mathbb{E}[Y^2] = \sigma^2$. By (0.9.63), and since $\varphi_{cY}(t) = \varphi_Y(ct)$ straight from the definition,

$$\varphi_{Z_n}(t) = \left[\varphi_Y\left(\frac{t}{\sigma\sqrt{n}}\right) \right]^n. \quad (0.9.68)$$

Take logarithms, which turns the power into a product and is the move that makes the whole thing work. Expand φ_Y to second order using (0.9.60) with $\mu_Y = 0$, then expand $\ln(1 + w) = w - \frac{1}{2}w^2 + \dots$ from Chapter 0.3:

$$\begin{aligned} \ln \varphi_{Z_n}(t) &= n \ln \left[1 - \frac{\sigma^2}{2} \frac{t^2}{\sigma^2 n} + O(n^{-3/2}) \right] \\ &= n \left[-\frac{t^2}{2n} + O(n^{-3/2}) \right] = -\frac{t^2}{2} + O(n^{-1/2}). \end{aligned} \quad (0.9.69)$$

Watch what happened, because it is the entire theorem. The quadratic term carried exactly one power of $1/n$, so multiplying by n left it *finite and unchanged*. Every higher term carried a higher power of $1/\sqrt{n}$, so multiplying by n still left it going to zero. The quadratic term is the unique survivor — not because we truncated, but because the scaling by $\sqrt{n}$ in (0.9.67) was chosen precisely to make it so. Hence

$$\varphi_{Z_n}(t) \longrightarrow e^{-t^2/2}, \quad (0.9.70)$$

and by Chapter 0.2's Gaussian integral with $a = \frac{1}{2}$, $b = it$, the function $e^{-t^2/2}$ is the characteristic function of the standard normal density $e^{-x^2/2}/\sqrt{2\pi}$ — as you can check by doing that integral. So the limit is a Gaussian, and it is a Gaussian because $e^{-t^2/2}$ is the fixed point of "square and rescale", which is what (0.9.68) does.

What is not proved here. (i) The remainder $O(n^{-3/2})$ needs control uniform in n ; doing it properly requires either a third moment or a truncation argument, and the honest general statement is Lindeberg's. (ii) Convergence of characteristic functions implies convergence of distributions — that is **Lévy's continuity theorem**, and we are quoting it. Neither gap affects the mechanism: the Gaussian appears because the quadratic term is the only one that survives the $\sqrt{n}$ scaling, and that part is fully derived above.

▼ Grind box — when the central limit theorem is false

The hypothesis $\sigma^2 < \infty$ is doing real work. Drop it and the conclusion fails, in an instructive way.

Take the Cauchy density $p(x) = \frac{1}{\pi(1+x^2)}$, whose tails fall as x^{-2} , so that $\int x^2 p \, dx$ diverges — no variance. Its characteristic function is $\varphi(t) = e^{-|t|}$ (a contour integral, quoted here). Then by (0.9.63) the mean of n samples has

$$\varphi_{\bar{X}_n}(t) = \left[e^{-|t|/n} \right]^n = e^{-|t|},$$

the characteristic function of a single Cauchy sample. **Averaging a million Cauchy observations gives you exactly the precision of one observation.** The $1/\sqrt{n}$ is gone, because (0.9.65) never applied: there was no variance to add. Note also that $e^{-|t|}$ is not differentiable at $t = 0$ — the failure of the mean to exist is visible as a corner in the characteristic function, exactly where (0.9.60) would have read it off.

This is not a pathology invented to embarrass the theorem. Cauchy tails arise from ratios of normal variables, from resonance line shapes (the Lorentzian of (0.9.25) is a Cauchy density), and in any setting where the largest single contribution is comparable to the sum of the rest. When you see a sample mean that refuses to stabilise as data accumulate, this is the first thing to suspect.

FAMILIAR GROUND

Equation (0.9.66) is the most expensive line of algebra in clinical research, and it is worth seeing that it is a *theorem* rather than a statistical convention. The standard error of a mean is $\sigma/\sqrt{n}$ because variances add for independent observations (0.9.65), which is true because a cross term $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ vanishes. That is the whole reason. No other exponent is available.

The consequences are brutal and entirely determined. A confidence interval has half-width proportional to $\sigma/\sqrt{n}$, so **halving the width of an interval costs four times the sample size**, and cutting it to a third costs nine times. For a two-arm trial comparing means with two-sided $\alpha = 0.05$ and 80% power, the required size per arm is

$$n = \frac{2(z_{0.975} + z_{0.80})^2 \sigma^2}{\delta^2} = \frac{15.7 \sigma^2}{\delta^2},$$

with $z_{0.975} = 1.960$ and $z_{0.80} = 0.842$. The detectable effect δ enters *squared*, which is (0.9.66) inverted: chasing an effect half the size costs four times the patients, and an effect a quarter the size costs sixteen times. Every argument you have ever had about whether a trial is powered for a realistic effect is an argument about that exponent, and the exponent is not negotiable.

Now the bridge, and it is an identity rather than an analogy. *Counting* is a sum of independent yes/no events, so (0.9.65) applies to it verbatim. One event either registers or does not, with probability p ; by (0.9.57) its variance is $\mathbb{E}[X^2] - \mu^2 = p - p^2$, since $X^2 = X$ for a variable taking only the values 0 and 1. Add M independent such events: the mean count is $N = Mp$ and the variance is $Mp(1 - p) \rightarrow N$ for small p . So **N counts carry a standard deviation of $\sqrt{N}$** and a relative precision $\sqrt{N}/N = 1/\sqrt{N}$ — derived, in three lines, from the same cross term vanishing. That single fact sets the noise floor of every measurement in physics — photon shot noise in a detector, the error bar on a bin of a collider histogram, the counting error on a radioactive sample, the sensitivity of a gravitational-wave interferometer. When a particle physicist demands 5σ before claiming a discovery, they are asking for an excess whose probability under the null is 2.9×10^{-7} — and the σ they are counting in is $\sqrt{N}$, from (0.9.65). When you compute the power of a trial, you use the same line with N replaced by patients.

So a discovery threshold at CERN and a sample-size calculation on a protocol sheet rest on identical algebra, derived above in four lines. The difference between the two fields is which quantity is expensive to increase.

△ WHY THIS ISN'T OBVIOUS

The delta is not a function, and pretending otherwise produces real damage. There is no assignment of a value to each point of the real line that does what (0.9.31) requires. The candidate — zero everywhere except one point — fails, because changing a function at a single point does not change any integral at all; that candidate has $\int \delta f = 0$ for every f . Nor can the value at the origin be "infinity", since infinity is not a real number and no arithmetic involving it would make the integral come out to $f(0)$ rather than something else. The delta only means anything *under an integral sign*, paired with a test function, and its type is a linear functional, not a function.

Treat it as a function anyway and you immediately generate two expressions that look harmless and are not. $\delta(0)$: setting $x = 0$ in (0.9.36) gives $\delta(0) = \frac{1}{2\pi} \int dk$, a divergent integral over all wavenumbers. $\delta(x)^2$: there is no product of distributions, and $\int \delta(x)^2 f(x) dx$ would have to be " $\delta(0)f(0)$ ", which is the previous problem again.

Both show up in quantum field theory, unavoidably and on the first page of any real calculation. A $\delta^4(p)$ enforcing momentum conservation gets squared when you square an amplitude, and the resulting $\delta^4(0)$ is the volume of spacetime — infinite, and correctly so, because you asked for a total probability rather than a rate per unit volume. Products of propagators at coincident points produce the same disease. In every case the appearance of $\delta(0)$ or δ^2 is not a mistake; **it is a signal that the question was posed in a limit that has not yet been taken carefully**, and the repair is a regulator — put the system in a box, smear the point, analytically continue the dimension — exactly as §5.3 inserted $e^{-\epsilon k^2}$, computed, and then removed it. Chapter 5.9 does nothing else for an entire chapter and calls it renormalisation.

8 · Worked examples

WORKED EXAMPLE 1 — THE GAUSSIAN, TRANSFORMED IN FULL

Compute $\tilde{f}$ for $f(x) = e^{-ax^2}$, $a > 0$, and verify $\Delta x \Delta k = \frac{1}{2}$ exactly.

Straight from the definition (0.9.12):

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2} e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2 - ikx} dx.$$

Complete the square, exactly as in Chapter 0.2:

$$-ax^2 - ikx = -a \left(x^2 + \frac{ik}{a}x \right) = -a \left(x + \frac{ik}{2a} \right)^2 - \frac{k^2}{4a}.$$

The second term is constant in x and comes out. The first is a Gaussian shifted by an *imaginary* amount, and shifting the contour off the real axis is the one step that needs a word: it is legal because e^{-az^2} is analytic and decays in the strip between the two contours, so the rectangle-contour argument of Chapter 0.4 gives zero for the connecting edges. (This is the same continuation Chapter 0.2's Problem 4 checked independently by solving a differential equation in k instead — two routes, one answer, which is how you know the continuation is sound.) With Chapter 0.2's $\int e^{-au^2} du = \sqrt{\pi/a}$:

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{\pi}{a}} e^{-k^2/4a} = \frac{1}{\sqrt{2a}} e^{-k^2/4a}.$$

A Gaussian transforms to a Gaussian, of reciprocal width. A large a (narrow in x) produces a small $1/4a$ in the exponent (broad in k), and vice versa.

The widths. The densities are $|f|^2 = e^{-2ax^2}$ and $|\tilde{f}|^2 \propto e^{-k^2/2a}$. Matching each against $e^{-u^2/2\sigma^2}$:

$$2a = \frac{1}{2(\Delta x)^2} \Rightarrow \Delta x = \frac{1}{2\sqrt{a}}, \quad \frac{1}{2a} = \frac{1}{2(\Delta k)^2} \Rightarrow \Delta k = \sqrt{a}.$$

$$\Delta x \Delta k = \frac{1}{2\sqrt{a}} \cdot \sqrt{a} = \frac{1}{2} \quad \text{for every } a. \quad \checkmark$$

Two things to notice. First, the answer is independent of a — squeeze the function and the transform spreads by precisely the reciprocal factor, which is the behaviour the figure in §6.5 measures numerically. Second, this is the equality case of (0.9.52), and §6.5 proved it is the *only* one. Chapter 0.2's Problem 4 got here first, by a completely different route, and said so.

WORKED EXAMPLE 2 — DIFFUSION, SOLVED BY TRANSFORMING, AND A DEBT REPaid

Solve $\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$ with all the material initially at the origin, $C(x, 0) = \delta(x)$.

Chapter 0.7 got the scaling $x \sim \sqrt{Dt}$ out of this equation by dimensional analysis: D has units of $\text{length}^2/\text{time}$, so the only length available is $\sqrt{Dt}$. That argument is correct and gives no constant and no shape. Here is the actual solution.

Transform in x only, holding t as a parameter. By (0.9.20) applied twice, $\partial_x^2 \rightarrow (ik)^2 = -k^2$, while the time derivative passes through the x -integral untouched:

$$\frac{\partial \tilde{C}(k, t)}{\partial t} = -Dk^2 \tilde{C}(k, t).$$

The partial differential equation has become, for each k separately, the first-order ordinary equation of Chapter 0.1 — rate proportional to amount. Its solution is the exponential, with a decay constant Dk^2 that depends on the mode:

$$\tilde{C}(k, t) = \tilde{C}(k, 0) e^{-Dk^2 t}.$$

The initial condition. By (0.9.31), the transform of δ is $\tilde{C}(k, 0) = \frac{1}{\sqrt{2\pi}} \int \delta(x) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}}$: a constant, the same at every wavenumber. *A point source contains every mode equally.* So

$$\tilde{C}(k, t) = \frac{1}{\sqrt{2\pi}} e^{-Dk^2 t},$$

and the physics is already visible without inverting anything: each mode decays at rate Dk^2 , so short-wavelength structure is erased fastest, quadratically fastest. That is what smoothing is.

Transform back with (0.9.14), using Chapter 0.2's Gaussian integral with $a = Dt$ and $b = ix$:

$$\begin{aligned} C(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-Dk^2 t} e^{ikx} dk = \frac{1}{2\pi} \sqrt{\frac{\pi}{Dt}} e^{-x^2/4Dt} \\ &= \frac{1}{\sqrt{4\pi Dt}} e^{-x^2/4Dt}. \end{aligned}$$

Read the answer. It is a Gaussian for every $t > 0$, with total integral 1 at all times (Chapter 0.2 again) — matter is conserved. Matching $e^{-x^2/4Dt}$ against $e^{-x^2/2\sigma^2}$ gives

$$\sigma(t) = \sqrt{2Dt}.$$

This **closes the loop Chapter 0.7 opened**. There, the $\sqrt{Dt}$ scaling came from dimensional analysis, and Problem 3 of that chapter said explicitly that the method could give the scaling but never the pure number in front. Here is the number: the distribution is Gaussian, its variance is $\langle x^2 \rangle = 2Dt$, and the constant is $\sqrt{2}$. Nothing was guessed. It also explains why the answer had to be Gaussian — the solution is the density of a sum of many independent random steps, and §7.5 says such a sum is Gaussian. The heat kernel and the central limit theorem are the same statement, which is why Chapter 5.8's path integral will be able to treat diffusion and quantum amplitudes with one piece of machinery.

One last reading: $C(x, t)$ is the Green's function of the diffusion operator, so by §4.2 the solution for *any* initial profile is the convolution $C(x, t) = \int C_0(y) C(x - y, t) dy$. Diffusion is convolution with a spreading Gaussian, which is precisely why repeated diffusion steps compose by adding variances.

9 • Your turn

PROBLEM 1 — THE SQUARE PULSE, THE SINC, AND A DIVERGENCE WORTH UNDERSTANDING

Let $f(x) = 1$ for $|x| \leq a$ and 0 otherwise. (a) Compute $\tilde{f}$. (b) Verify Plancherel (0.9.17) for it, given $\int_{-\infty}^{\infty} (\sin u/u)^2 du = \pi$. (c) Compute Δx . (d) Attempt Δk , and explain what you find in terms of (0.9.52).

– Solution

(a) Directly from (0.9.12):

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} \cdot \frac{2 \sin(ka)}{k} = \sqrt{\frac{2}{\pi}} \frac{\sin(ka)}{k},$$

using the same $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ step as (0.9.3). This is the **sinc** function: a central lobe of half-width π/a , then side lobes of alternating sign whose envelope decays only as $1/k$.

(b) $\int |f|^2 dx = 2a$. On the other side, substituting $u = ka$,

$$\int |\tilde{f}|^2 dk = \frac{2}{\pi} \int \frac{\sin^2(ka)}{k^2} dk = \frac{2a}{\pi} \int \frac{\sin^2 u}{u^2} du = \frac{2a}{\pi} \cdot \pi = 2a. \checkmark$$

(c) $|f|^2 = f$, so

$$(\Delta x)^2 = \frac{\int_{-a}^a x^2 dx}{\int_{-a}^a dx} = \frac{2a^3/3}{2a} = \frac{a^2}{3}, \quad \Delta x = \frac{a}{\sqrt{3}}.$$

(d) Now attempt the same in k . The numerator is

$$\int_{-\infty}^{\infty} k^2 |\tilde{f}|^2 dk = \frac{2}{\pi} \int_{-\infty}^{\infty} \sin^2(ka) dk,$$

because the k^2 cancels the $1/k^2$ exactly. The integrand no longer decays at all: it oscillates between 0 and 1 forever, and the integral **diverges**. So $\Delta k = \infty$ and $\Delta x \Delta k = \infty \geq \frac{1}{2}$ — the theorem holds, in the loudest possible way.

This is not a technicality; it is the theorem telling you something true. A hard edge is infinitely sharp, and by §6.2 the spread in k is the size of f' — which for a step is a delta, with infinite norm. *A discontinuity costs infinite bandwidth.*

Truncating at $|k| \leq K$ shows how it diverges. Then $\int_{-K}^K k^2 |\tilde{f}|^2 dk \approx 2K/\pi$ (the $\sin^2$ averages to $\frac{1}{2}$), so $(\Delta k)^2 \approx K/\pi a$ and

$$\Delta x \Delta k \approx \frac{a}{\sqrt{3}} \sqrt{\frac{K}{\pi a}} = \sqrt{\frac{aK}{3\pi}} \rightarrow \infty.$$

This is what the §6.5 figure is doing on the square-pulse preset: the readout is tracking the sampling cutoff K , not converging to anything. Double the number of grid points and Δk rises by a factor of $\sqrt{2}$ — measured values 3.23, 4.64, 6.61, 9.38, 13.29 for $N = 128, \dots, 2048$, ratios 1.44, 1.42, 1.42, 1.42. (The discrete pulse has its own sampling corrections on top, so the number does not match the continuum formula exactly; the point is that there is no limit for it to match.) Worth knowing where a number comes from before you trust it.

A finite honest statement. 90.28% of the power lies in the central lobe $|k| < \pi/a$ (numerically, from $\frac{1}{\pi} \int_{-\pi}^{\pi} (\sin u/u)^2 du$). If you use the central-lobe half-width π/a as the bandwidth, the product is $(a/\sqrt{3})(\pi/a) = \pi/\sqrt{3} = 1.814$, comfortably above $\frac{1}{2}$. Different width measure, different constant, same conclusion — and it is the last 10% of the power, out in the tails, that makes the variance-based measure blow up.

PROBLEM 2 — THE SCALING LAW FOR THE DELTA, IN THREE DIMENSIONS TOO

From the defining property (0.9.31) alone, prove $\delta(ax) = \delta(x)/|a|$ for real $a \neq 0$. Then deduce $\delta(-x) = \delta(x)$, find the analogous rule for $\delta^3(\mathbf{ar})$, and use the result to evaluate $\int \delta(x^2 - c^2)f(x) dx$ for $c > 0$.

– Solution

The scaling law. Two distributions are equal when they give the same number on every test function, so test both sides. Substitute $u = ax$, $dx = du/a$.

For $a > 0$ the limits keep their order:

$$\int_{-\infty}^{\infty} \delta(ax)f(x)dx = \int_{-\infty}^{\infty} \delta(u)f(u/a)\frac{du}{a} = \frac{f(0)}{a}.$$

For $a < 0$ the substitution sends $x = -\infty \mapsto u = +\infty$, so the limits arrive reversed and swapping them costs a sign:

$$\int_{+\infty}^{-\infty} \delta(u)f(u/a)\frac{du}{a} = -\frac{f(0)}{a} = \frac{f(0)}{|a|}.$$

Both cases give $f(0)/|a|$, which is what $\int [\delta(x)/|a|] f dx$ gives. Equal on every test function, hence equal. ■

Consequence. Put $a = -1$: $\delta(-x) = \delta(x)$, so the delta is even — which had better be true, since (0.9.36) is manifestly even in x .

Three dimensions. $\delta^3(\mathbf{ar}) = \delta(ax)\delta(ay)\delta(az)$, and each factor contributes $1/|a|$:

$$\delta^3(\mathbf{ar}) = \frac{\delta^3(\mathbf{r})}{|a|^3}, \quad \text{and in } n \text{ dimensions } \frac{1}{|a|^n}.$$

The exponent is the dimension, because the delta's units are one over a volume. In Chapter 5.6 the deltas are four-dimensional and this is how their Jacobians are tracked.

The composite argument. $g(x) = x^2 - c^2$ vanishes at $x = \pm c$, and near each zero it is approximately linear (Chapter 0.1): $g(x) \approx g'(\pm c)(x \mp c)$ with $g'(\pm c) = \pm 2c$. The delta only sees neighbourhoods of the zeros, so it splits into one term per zero, each rescaled by the local slope via the law just proved:

$$\begin{aligned} \delta(x^2 - c^2) &= \frac{\delta(x - c)}{|2c|} + \frac{\delta(x + c)}{|-2c|}, \\ \implies \int \delta(x^2 - c^2)f(x)dx &= \frac{f(c) + f(-c)}{2c}. \end{aligned}$$

The general rule, $\delta(g(x)) = \sum_i \delta(x - x_i)/|g'(x_i)|$ over the simple zeros x_i , is this argument run once. It is used on every page of relativistic kinematics, where $\delta(p^2 - m^2)$ puts a particle on its mass shell and

the $1/|g'|$ becomes the factor $1/2E$ in the phase-space measure.

PROBLEM 3 — GAUSSIANS ARE CLOSED UNDER ADDITION

Let $X \sim N(0, \sigma_1^2)$ and $Y \sim N(0, \sigma_2^2)$ be independent. Using the convolution theorem rather than a direct integral, show that $X + Y$ is Gaussian with variance $\sigma_1^2 + \sigma_2^2$. Then say what would have gone wrong without independence.

– Solution

Step 1 — the characteristic function of a Gaussian. With $p(x) = e^{-x^2/2\sigma^2}/\sigma\sqrt{2\pi}$, Chapter 0.2's $\int e^{-ax^2+bx} dx = \sqrt{\pi/a} e^{b^2/4a}$ with $a = 1/2\sigma^2$, $b = it$ gives

$$\varphi(t) = \frac{1}{\sigma\sqrt{2\pi}} \sqrt{2\pi\sigma^2} e^{(it)^2\sigma^2/2} = e^{-\sigma^2 t^2/2}.$$

Step 2 — multiply. The density of $Z = X + Y$ is $p_X * p_Y$ by (0.9.62), so by (0.9.63) the characteristic functions multiply:

$$\varphi_Z(t) = e^{-\sigma_1^2 t^2/2} e^{-\sigma_2^2 t^2/2} = e^{-(\sigma_1^2 + \sigma_2^2)t^2/2}.$$

Step 3 — recognise it. That is Step 1 again with $\sigma^2 = \sigma_1^2 + \sigma_2^2$. Since the characteristic function determines the density uniquely (the transform is invertible, §7.2), $Z \sim N(0, \sigma_1^2 + \sigma_2^2)$. ■

Compare the effort: convolving two Gaussians directly means completing a square inside a double exponential. In Fourier space, adding two exponents. This is (0.9.28) earning its keep, and it is the reason the Gaussian family is closed under addition at all — the transform turns convolution into multiplication, and the *exponential of a quadratic* is exactly the shape that is closed under multiplication, because quadratics add.

Without independence. Step 2 is the only place it was used, and it fails completely: p_Z is not $p_X * p_Y$ unless the joint density factorises, so the characteristic functions do not multiply. Correlated Gaussians still sum to a Gaussian (that is a separate fact about the joint normal distribution), but the variance is $\sigma_1^2 + \sigma_2^2 + 2\text{cov}(X, Y)$, from (0.9.64) with the cross term retained. Positively correlated errors add up faster than $\sqrt{n}$; this is why correlated measurements do not average down, and why cluster randomisation costs more patients than it looks like it should.

PROBLEM 4 — SHORT PULSES AND NARROW LINES

(a) From (0.9.55), show that a Gaussian pulse whose intensity has full-width-at-half-maximum Δt has a frequency content of FWHM $\Delta \nu = 2 \ln 2 / \pi \Delta t = 0.441 / \Delta t$. Evaluate the required spectral width, in nanometres, for a 10 fs pulse centred at 800 nm. (b) An excited state decays with lifetime τ , so its amplitude behaves as $a(t) = e^{-i\omega_0 t} e^{-t/2\tau}$ for $t > 0$ and 0 before. Transform it, show the intensity spectrum is a Lorentzian, and find the linewidth. Evaluate for $\tau = 1$ ns.

– Solution

(a) For a Gaussian, (0.9.55) gives $\Delta t \Delta \omega = \frac{1}{2}$ in standard deviations, i.e. $\sigma_t \sigma_\omega = \frac{1}{2}$ (the pulse is a function of t , its transform a function of ω ; nothing changes but the names). A Gaussian intensity $e^{-u^2/2\sigma^2}$ has FWHM $2\sqrt{2 \ln 2} \sigma$, from solving $e^{-u^2/2\sigma^2} = \frac{1}{2}$. So

$$\text{FWHM}_t \cdot \text{FWHM}_\omega = 8 \ln 2 \sigma_t \sigma_\omega = 8 \ln 2 \cdot \frac{1}{2} = 4 \ln 2,$$

and converting the angular frequency to ordinary frequency, $\omega = 2\pi\nu$:

$$\Delta t \Delta \nu = \frac{4 \ln 2}{2\pi} = \frac{2 \ln 2}{\pi} = 0.4413.$$

This is the *transform limit*: a pulse cannot be shorter than this for its bandwidth, and a pulse that achieves it is called transform-limited.

Numbers. $\Delta t = 10$ fs gives $\Delta \nu = 0.4413/10^{-14} \text{ s} = 4.41 \times 10^{13} \text{ Hz} = 44.1 \text{ THz}$. Converting to wavelength via $\nu = c/\lambda$, so $|d\nu| = c |d\lambda|/\lambda^2$ (Chapter 0.1's logarithmic derivative):

$$\Delta \lambda = \frac{\lambda^2 \Delta \nu}{c} = \frac{(800 \times 10^{-9})^2 \cdot 4.41 \times 10^{13}}{3.00 \times 10^8} = 9.4 \times 10^{-8} \text{ m} = 94 \text{ nm}.$$

A 10 fs pulse at 800 nm must span roughly 94 nm of spectrum — more than a tenth of its own centre wavelength, and a large fraction of the visible range. This is why ultrafast lasers need broadband gain media and why "monochromatic ultrashort pulse" is a contradiction, not an engineering challenge.

(b) Write the carrier as $e^{+i\omega_0 t}$, so that with our forward kernel $e^{-i\omega t}$ the spectral line lands at $+\omega_0$. Transform with (0.9.12), integrating only over $t > 0$ where $a \neq 0$; the exponent collects to $-i(\omega - \omega_0)t - t/2\tau$ and the integral is elementary:

$$\begin{aligned} \tilde{a}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-[i(\omega - \omega_0) + \frac{1}{2\tau}]t} dt \\ &= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{i(\omega - \omega_0) + \frac{1}{2\tau}}. \end{aligned}$$

(The evaluation at the upper limit vanishes because of the $e^{-t/2\tau}$ factor — the decay is what makes the transform exist at all.)

The measured intensity is $|\tilde{a}|^2$, and the modulus squared of a reciprocal is the reciprocal of the modulus squared:

$$|\tilde{a}(\omega)|^2 = \frac{1}{2\pi} \cdot \frac{1}{(\omega - \omega_0)^2 + \left(\frac{1}{2\tau}\right)^2}.$$

A Lorentzian — the identical shape as the driven-oscillator response (0.9.25), which is not a coincidence: an excited state *is* a damped oscillator, with $\gamma = 1/2\tau$. Its half-width at half maximum is $1/2\tau$, so the full width is

$$\Delta\omega = \frac{1}{\tau}, \quad \Delta E = \hbar\Delta\omega = \frac{\hbar}{\tau}.$$

Numbers. For $\tau = 1 \text{ ns}$: $\Delta\omega = 10^9 \text{ s}^{-1}$, so $\Delta\nu = \Delta\omega/2\pi = 159 \text{ MHz}$, and $\Delta E = 1.055 \times 10^{-25} \text{ J} = 0.66 \text{ } \mu\text{eV}$. Against an optical transition at 500 nm ($\nu_0 = 6.00 \times 10^{14} \text{ Hz}$) that is a fractional width of 2.7×10^{-7} — spectacularly sharp, and yet not zero, and not zero for a reason that has nothing to do with the apparatus.

The point. The relation $\Delta E \tau \sim \hbar$ is usually presented as a fourth Heisenberg relation, "energy–time uncertainty", with an air of mystery. It is (0.9.52) applied in the time domain, plus $E = \hbar\omega$. A state that does not last forever does not have a sharp energy, for the same reason a note that does not last forever does not have a sharp pitch. No measurement anywhere.

You built the Fourier basis as an **orthonormal basis**, with the orthogonality (0.9.4) proved by direct integration and the coefficients recognised as Chapter 0.5's inner products. You took $L \rightarrow \infty$ honestly and got the **transform pair**, proved **Plancherel** and read it as the statement that the transform is **unitary** — a rotation in function space. You showed that the basis **diagonalises the derivative**, which is the reason the subject exists, and turned two differential equations into division. You derived the **convolution theorem** and saw that a linear system's response is a Green's function convolution, hence a product of transfer functions — hence a product of propagators. You gave the **Dirac delta** a definition it can survive, derived its Fourier representation with a regulator, and recognised it as Chapter 0.5's completeness relation in continuous disguise. You proved the **bandwidth theorem** $\Delta x \Delta k \geq \frac{1}{2}$ from Cauchy–Schwarz, showed the Gaussian is its unique minimiser, and measured that number live. And you found the **characteristic function** to be a Fourier transform, which made the central limit theorem a statement about which Taylor term survives a rescaling.

Where this gets spent.

- **The Fourier basis and Plancherel** — Chapter 4.3 (where completeness is finally proved and the position and momentum representations become two bases for one Hilbert space), Chapter 4.4 (where unitarity is conservation of probability), Chapter 5.3 (where a quantum field is expanded in exactly (0.9.14), one harmonic oscillator per k).
- **Diagonalising the derivative** — Chapter 5.6, where every propagator is (0.9.24) with ω replaced by a four-momentum, and where momentum space is the only place the calculations are tractable.
- **Convolution and Green's functions** — Chapter 5.6 again: the Green's function of a wave operator is the *propagator*, and Feynman diagrams are products precisely because convolution in position space is multiplication in momentum space (0.9.28).
- **The delta** — Chapter 4.3 (continuum normalisation $\langle x|y \rangle = \delta(x - y)$), Chapter 5.3 (equal-time commutators), Chapter 5.9 (where $\delta(0)$ and δ^2 appear and force regularisation).
- **The bandwidth theorem** — Chapter 4.6, which adds $p = \hbar k$ and nothing else.
- **Probability, variance, the CLT** — Chapter 4.8 (density matrices and measurement statistics), Chapter 5.9 (where the renormalisation group is a central-limit theorem for fluctuations at successive scales — the same "only one term survives the rescaling" argument as (0.9.69)).

And that closes Part 0. Nine chapters, one accumulating toolkit, no physics yet. You have the **linear approximation** (0.1), the **integral** as an accumulation with the Gaussian done three ways (0.2), the **series expansion** with control of its error (0.3), the **vector space** and the linear map (0.4), the **spectral theorem** in an inner-product space (0.5), the **total derivative** and the Jacobian (0.6), the **field theorems** that turn interior cancellation into boundary integrals (0.7), the **oscillator** and the differential equations around it (0.8), and now the **Fourier transform** that makes all of them commute with each other.

Chapter 1.1 begins the physics. What happens there is not that you start using one of these tools. It is that the first serious question in mechanics — *why do objects move along the paths they do?* — turns

out to require the linear approximation, the integral, the expansion, and the variational idea *all at once, in the first three pages*. That is the point at which Part 0 stops looking like a long detour and starts looking like the only sensible way in.